EXERCISE 03

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October 28, 2026

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1 Introduction

Solution of Exercise 03 on the randomised perceptron learning rule.

2 Methodology

2.1 Randomised learning rule

For each training set, the student's initial couplings are independently drawn from a Gaussian distribution $\mathcal{N}(0, 1)$. Each example is presented sequentially. Only when the student classification differs from the teacher label are the couplings updated according to

$$J_i(t+1) = J_i(t) + \frac{1}{\sqrt{N}}\,\sigma_T^\mu \xi_i^\mu (1 + r_i), \qquad r_i \sim \mathcal{N}(0, 50). \tag{1}$$

Thus the Gaussian perturbation has *variance* 50, or standard deviation $\sqrt{50}$. A new independent value r_i is drawn for each coupling at each update. After every full cycle (epoch), the training error is evaluated on the entire training set. Training stops when $\epsilon_t = 0$.

2.2 Performance measures

The test error of each trained student is computed on $P_{\text{test}} = 1000$ new examples, independent of the training set. For each pair (N, P), the whole training and testing procedure is repeated $R = 1000$ times. The reported mean error and its statistical uncertainty are

$$\langle \epsilon \rangle = \frac{1}{R} \sum_{k=1}^{R} \epsilon_k, \qquad \text{SEM} = \frac{s_\epsilon}{\sqrt{R}}, \qquad s_\epsilon^2 = \frac{1}{R-1} \sum_{k=1}^{R} (\epsilon_k - \langle \epsilon \rangle)^2. \tag{2}$$

All plotted error bars represent ± 1 standard error of the mean (SEM).

3 Results

3.1 Generalization error for $N = 20$

All seven training-set sizes were investigated using 1000 repetitions. The averaged test errors and their SEMs are reported in Table 1.

Table 1: Average test error for $N = 20$ over 1000 repetitions.

P	$\langle \epsilon \rangle$	SEM
1	0.491181	0.002410
10	0.415094	0.002137
20	0.349457	0.002117
50	0.212613	0.001450
100	0.119193	0.000978
150	0.080899	0.000718
200	0.060460	0.000542

The generalization error decreases as the number of training examples grows: it is close to 0.5 for $P = 1$ and falls to approximately 0.0605 for $P = 200$. The corresponding dependence on P is shown in Figure 1.

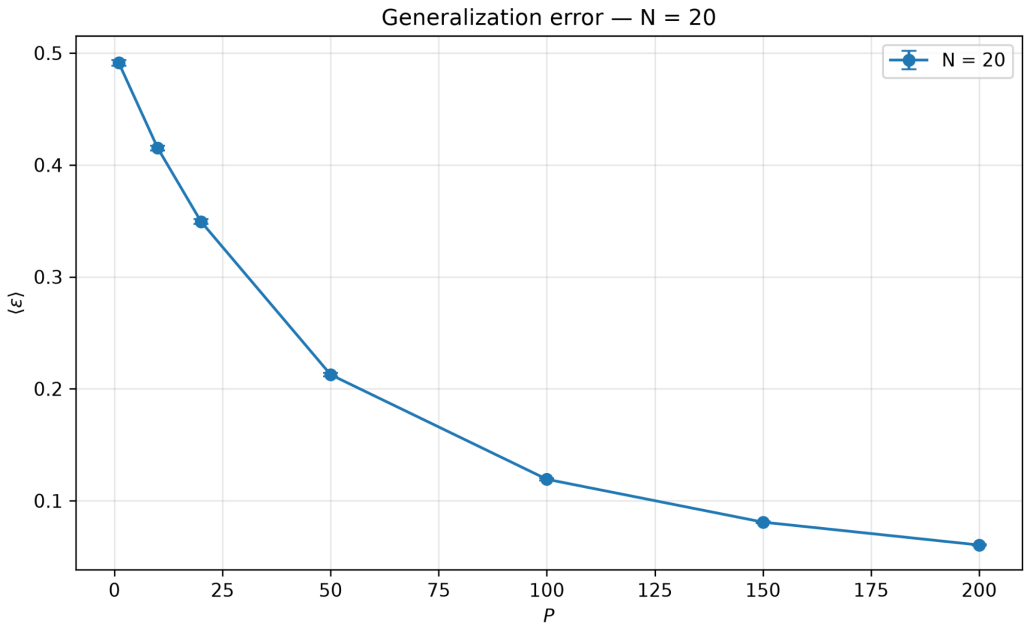


Figure 1: Average generalization error versus P for $N = 20$. Error bars show ± 1 SEM over 1000 repetitions.

3.2 Generalization error for $N = 40$

The same procedure was carried out for $N = 40$ and the seven prescribed values of P. The results are given in Table 2.

Table 2: Average test error for $N = 40$ over 1000 repetitions.

P	$\langle\epsilon\rangle$	SEM
1	0.495266	0.001685
2	0.488002	0.001729
20	0.417927	0.001580
40	0.347605	0.001513
100	0.213136	0.001071
200	0.120012	0.000732
300	0.082512	0.000527

Again, the test error decreases with increasing P, from approximately 0.5 for $P = 1$ to 0.0825 at $P = 300$.

3.3 Comparison between $N = 20$ and $N = 40$

Figure 2 shows both datasets against the number of examples P. For a fixed P in the investigated range, the system with $N = 40$ generally exhibits a higher test error than the system with $N = 20$. Consequently, the absolute number of examples alone does not explain the observed learning curves.

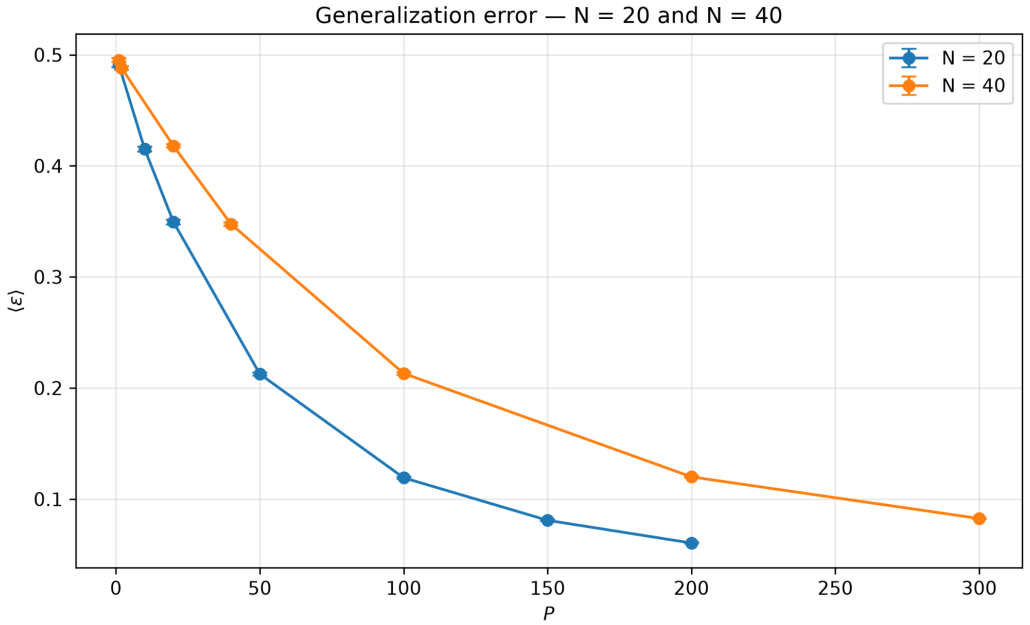


Figure 2: Average generalization error versus P for $N = 20$ and $N = 40$. Error bars show ± 1 SEM.

3.4 Data superposition

To compare systems with different numbers of couplings, the horizontal axis is rescaled using

$$\alpha = \frac{P}{N}. \tag{3}$$

The variable α measures the number of training examples per synaptic coupling. When the error is plotted against α, the two datasets nearly superimpose, as shown in Figure 3.

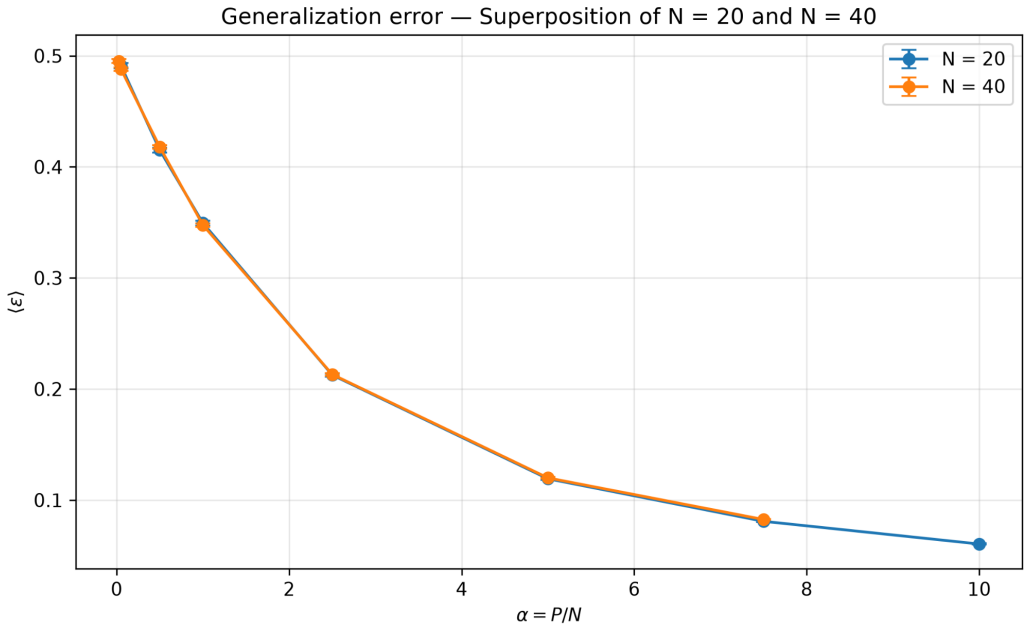


Figure 3: Approximate superposition of the average generalization error as a function of $\alpha = P/N$ for $N = 20$ and $N = 40$. Error bars show ± 1 SEM.

This agreement suggests that generalization performance is primarily controlled by P/N rather than by P alone. The superposition is approximate rather than mathematically exact. The observed residual differences are compatible with statistical fluctuations and possible finite-size effects; so the average generalization error approximately scales with P/N.

3.5 Dependence on noise variance (additional study)

The prescribed learning rule has perturbation variance $\sigma_r^2 = 50$. To investigate sensitivity to randomised updates, the $N = 40$ study was also performed with $\sigma_r^2 = 0$, 0.01, 1, 5, 20, 200, and 500, using

$$r_i \sim \mathcal{N}(0, \sigma_r^2). \tag{4}$$

The student initialisation, teacher, learning-rate factor $1/\sqrt{N}$, and training-set sizes remained unchanged. All reported curves are averages over 1000 repetitions, with error bars equal to ± 1 SEM.

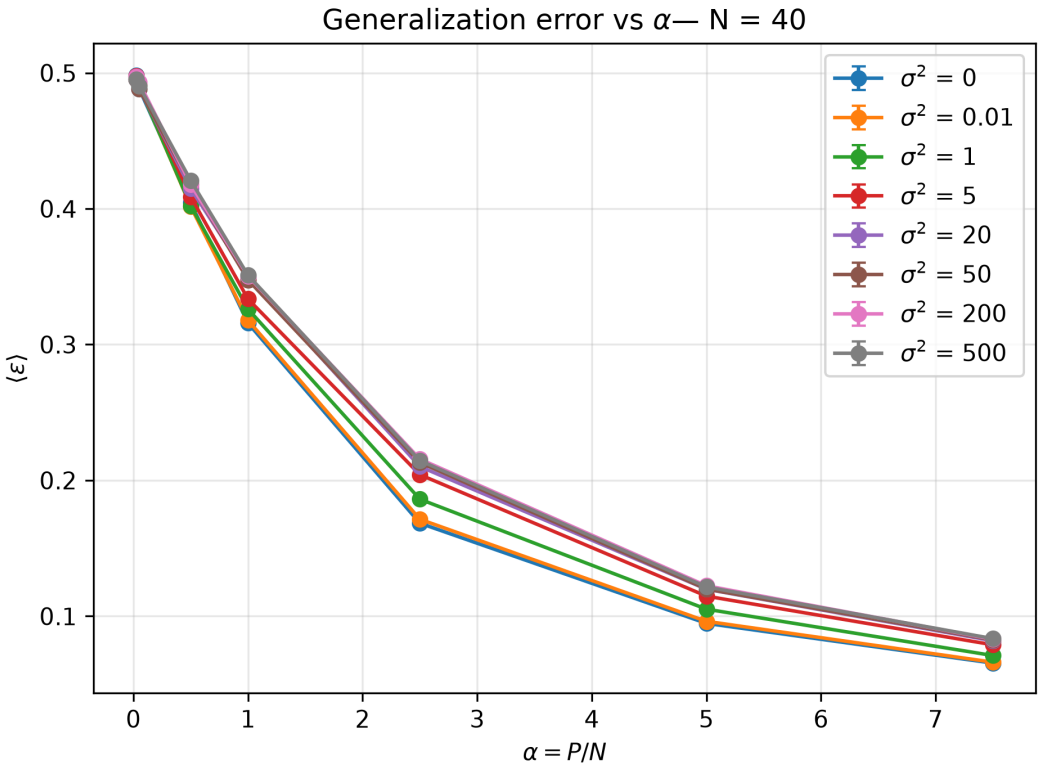


Figure 4: Average generalization error versus $\alpha = P/N$ for $N = 40$ at $\sigma_r^2 = 0$, 0.01, 1, 5, 20, 50, 200, and 500. Error bars show ± 1 SEM over 1000 independent repetitions.

Figure 4 shows that the error generally rises as the perturbation variance increases, especially at intermediate α. The effect becomes less pronounced for the largest variances considered; moreover for $\alpha \sim 0$ the generalization errors are all approximately equal to 0.5.

3.5.1 Execution times

Table 3 records the wall-clock times for each value of the variance in this $N = 40$ study.

Table 3: Approximate wall-clock times for a complete sweep of seven P values and 1000 repetitions at $N = 40$ for different variances.

σ_r^2	$N = 40$
0	3 s
0.01	3 s
1	4 s
5	5 s
20	16 s
50	36 s
200	135 s
500	327 s

The time required for convergence during training increases at large perturbation variance, and it's approximately equal and small for variances close to or equal to zero.
A low variance corresponds to a smaller generalization error and faster execution time, but explores the version space in a certain (almost deterministic) path
Higher variance introduces stronger stochasticity into the learning dynamics, potentially allowing the algorithm to explore different regions of the version space. Increasing the variance over a

certain value (in this case 50) gives approximately the same results but requires more computation time.

4 Conclusions

The randomised perceptron achieves zero training error for all investigated training sets, while its average test error decreases as the number of examples increases. The rescaling $\alpha = P/N$ gives a good approximate superposition of the $N = 20$ and $N = 40$ results for the reference variance $\sigma_r^2 = 50$. The supplementary variance scan for $N = 40$ shows that stronger random updates generally increase the generalization error, with a diminishing change at high variance. They can also substantially increase the time needed to reach zero training error.